

Guru Gobind Singh Indraprastha University



B. Tech. IT – VIIth Sem

Project Synopsis

on the topic of

**Vision Store: Retail Analytics from CCTV Footage Using
Computer Vision and POS Data**

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1. Introduction

Physical retail stores already have a rich source of behavioural data installed on their ceilings: CCTV cameras. In most stores, however, this footage is used only for security and is reviewed after an incident. Store managers, on the other hand, need answers to operational questions every day: How many people entered the store? Which sections attract the most visitors? How long do customers stay in each section? How many people are waiting at the billing counter right now? How many visitors actually purchased something? Online businesses answer these questions instantly through web analytics, but brick-and-mortar retailers rarely have an equivalent.

Recent progress in deep-learning-based object detection and multi-object tracking makes it possible to convert raw video into structured information. Modern detectors such as the YOLO family can locate people in real time, and tracking algorithms can follow each person across frames. When these outputs are mapped onto a store floor plan, they can be turned into meaningful business events such as entry, zone visit, dwell time, billing-queue join and exit.

Vision Store is a retail analytics system that converts CCTV footage into actionable store intelligence. The system detects and tracks people, separates staff from customers, maps movement onto store zones, and generates a compact stream of events instead of raw video or coordinates. These events are sent to a backend service, stored in a database, and combined with Point-of-Sale (POS) transaction data to compute metrics such as footfall, conversion rate, customer funnel, zone heatmaps, queue depth and abnormal activity. The results are presented on a near-real-time dashboard.

Two design principles distinguish this project. First, it is **privacy-first**: the system does not perform facial recognition, does not store biometric templates, and assigns only anonymous, session-scoped visitor identifiers; raw video stays within the store and only anonymous events leave the vision pipeline. Second, it is **measurable**: the POS transactions are used as an independent ground truth, so the accuracy of the vision-based funnel can be quantified rather than merely demonstrated. The central research question of the project is: how accurately can a privacy-preserving, CCTV-only pipeline estimate retail funnel metrics, when validated against POS records and manually annotated ground truth?

2. Objectives

- To design and implement an end-to-end pipeline that converts CCTV video into structured retail events (entry, zone enter/exit, dwell, queue join, exit).
- To detect and track people reliably and map their positions onto store zones using zone polygons and a floor-plane (homography) transformation.
- To distinguish store staff from customers so that staff activity does not distort visitor and dwell-time analytics.

- To build a backend service and database that ingest events and compute metrics, customer funnel, zone heatmap, queue depth and anomaly alerts, and to present them on a live dashboard.
- To correlate POS transactions with billing-zone visits and use POS records as ground truth for validating the estimated conversion rate.
- To benchmark alternative detectors and trackers and evaluate the system using standard measures such as precision, recall, mAP, MOTA, IDF1, HOTA, counting error and dwell-time error.
- To adopt a privacy-by-design approach: no facial recognition, no stored biometric data, anonymous identifiers and minimal data leaving the vision pipeline.

3. Feasibility Study

The feasibility of this project is high due to the following reasons:

3.1 Technical Feasibility

- The project relies entirely on mature, open-source tools. No specialised hardware is required; a standard laptop or desktop is sufficient, and a GPU (local or a free cloud notebook) speeds up model inference and benchmarking.
- Pre-trained person detectors are freely available, so no large-scale model training is needed. The contribution lies in integration, calibration, evaluation and analytics.
- The system is modular (vision pipeline, event layer, backend, dashboard), so each stage can be developed, tested and replaced independently.
- Video can also be replayed from recorded files, allowing repeatable experiments without a live camera feed.

Layer	Tools / Technologies (indicative)
Detection	YOLO-family detector (e.g., YOLOv8 / YOLO11), COCO-pretrained, person class
Tracking	ByteTrack, BoT-SORT and OC-SORT (to be benchmarked against each other)
Vision processing	Python, OpenCV, NumPy (zone polygons, homography, point-in-polygon tests)
Backend	FastAPI with Pydantic schemas; SQLite (upgradable to PostgreSQL / DuckDB)
Dashboard	Web dashboard with live refresh (polling / server-sent events)
Evaluation	Manual annotation of sample footage, MOT metrics, POS records, unit tests

3.2 Operational Feasibility

- **Time Frame:** The project can be completed in about one semester (roughly 16 weeks) in the phases shown below.
- **Data Requirements:** Store CCTV footage and the corresponding POS transaction export (CSV) are used as the primary data. Public benchmarks such as MOT17/MOT20 can be used to validate tracker behaviour independently of the store. Where real footage is limited, a synthetic traffic simulator with known ground truth can be used and will be clearly labelled as simulated in all results.
- **Implementation Steps:** The pipeline (acquisition → detection → tracking → zone mapping → events → analytics → evaluation) is modular and can be tested at each stage.
- **Experiment Flexibility:** Detectors, trackers, confidence thresholds, zone layouts and POS-matching windows can be varied quickly to study their effect on accuracy.

3.3 Research Feasibility

- **Active Research Domain:** Person detection, multi-object tracking and retail analytics are widely studied areas with well-defined benchmarks and metrics, which makes fair evaluation possible.
- **Measurable Outcomes:** Availability of POS data provides a rare, free source of ground truth for retail vision systems, enabling a quantitative study of estimation error.
- **Academic Relevance:** Known limitations such as occlusion, crowding, camera angle, lighting and staff/customer ambiguity can be analysed and reported, creating scope for identifying research gaps.
- **Future Scope:** The system can be extended with a natural-language assistant that answers store-manager queries through deterministic analytics tools, shelf-level intelligence (pick-up and put-back detection, out-of-stock alerts), and optimised edge deployment on low-power devices.

4. Need & Significance

- **Visibility into Physical Retail:** Unlike e-commerce, physical stores have little insight into visitor-to-buyer conversion. Vision Store fills this gap using infrastructure that already exists.
- **Cost Efficiency:** By reusing existing CCTV cameras, the system avoids the cost of dedicated people-counting sensors, making analytics accessible to small and medium retailers.
- **Operational Decisions:** Queue-depth alerts, zone heatmaps and traffic anomalies help managers open extra billing counters, reposition staff and improve store layout and product placement.

- **Privacy-Conscious Design:** Processing video into anonymous events, without facial recognition or stored biometric data, follows data-minimisation principles such as those emphasised in India's Digital Personal Data Protection Act, 2023.
- **Validated Accuracy:** Comparing vision-based estimates with POS records and manual annotations shows how reliable such systems actually are, which is often missing in demonstration-only projects.
- **Educational Value:** The project integrates computer vision, geometry, backend engineering, databases and evaluation methodology in one real-world application.

5. Methodology

The project development is carried out in the following steps:

1. **Data Acquisition:** Collect CCTV footage from one or more camera views and the matching POS transaction records. Record camera identifiers, timestamps and store layout.
2. **Preprocessing and Calibration:** Sample and resize frames, define store zones (e.g., entrance, product sections, billing) as polygons, and estimate a homography between the image and the floor plane where required.
3. **Person Detection:** Run a pre-trained YOLO-family detector restricted to the person class to obtain bounding boxes; the foot point of each box is used as the person's floor position.
4. **Multi-Object Tracking:** Associate detections across frames to obtain persistent track IDs. Compare ByteTrack, BoT-SORT and OC-SORT. Any appearance-based association is short-term, session-scoped and non-biometric.
5. **Role Classification:** Separate staff from customers using combined cues such as staff-only zones, unusually long dwell, movement pattern and, optionally, a small classifier for uniforms, rather than colour thresholds alone.
6. **Backend and Analytics:** Ingest events through a REST API into a database and compute footfall, dwell time, funnel stages, zone heatmap, queue depth and abandonment, and anomaly alerts. Match POS transactions to recent billing-zone visitors within a time window to estimate conversion.
7. **Evaluation and Documentation:** Compare outputs with manual ground truth and POS records, analyse errors and trade-offs, and document findings.

Methodology Flowchart

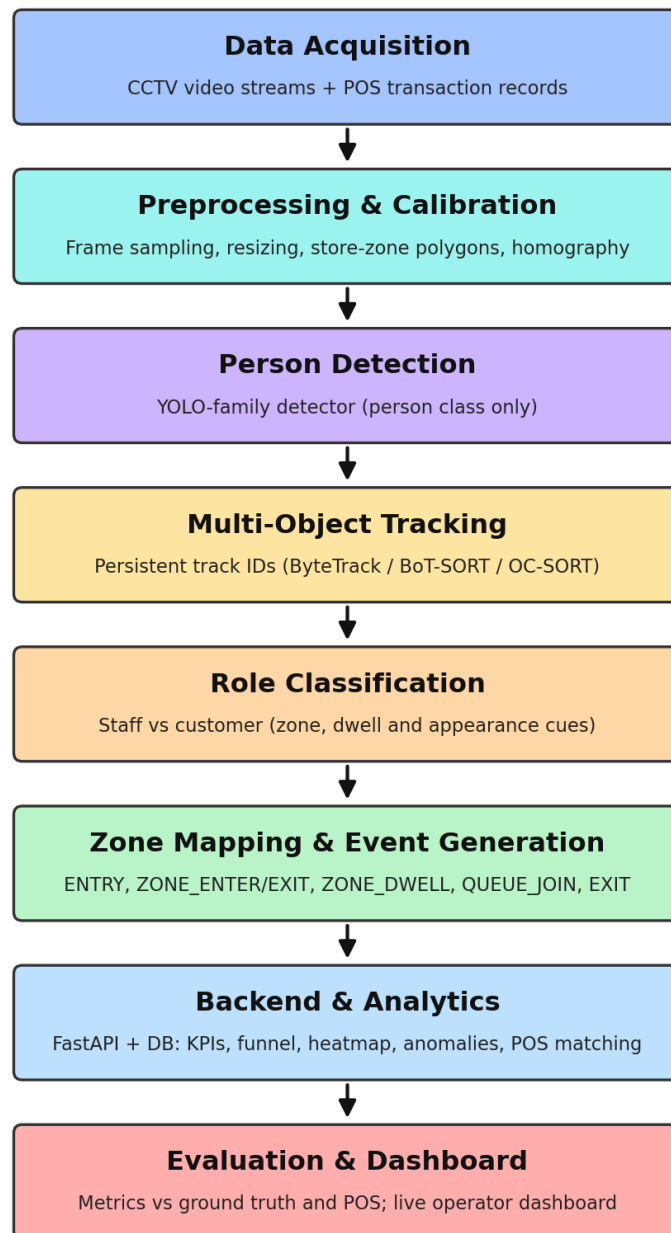


Figure 1: Vision Store methodology flowchart

Evaluation Plan

Component	Metrics	Reference
Person detection	Precision, recall, mAP	Manually annotated frames
Footfall counting	Mean absolute / percentage error	Manual entry count
Staff vs customer	Precision, recall, F1	Labelled tracks
Conversion estimate	Absolute error vs actual purchases	POS transaction records

Expected Outcomes

- A working prototype that turns CCTV video into store events, analytics and a live dashboard.
- A benchmark comparison of detectors and trackers on store footage.
- A quantified accuracy report of vision-based footfall, dwell time and conversion estimates against ground truth and POS data.
- A documented privacy-by-design approach and a discussion of limitations and future work.

6. Bibliography

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